

NLP WI2026, UCSD: Midterm Exam Topics

- The exam will include multiple choice, short answer, and long answer questions.
- For long answers, show your work for partial credit.
- Keep responses concise unless otherwise specified.
- Questions for will revolve around content in the lectures, and assignments.

Topics

Below is the list of topics that will be covered on the midterm, which is the bulk of the course material leading up to the midterm.

- **Intro to Neural NLP**
 - **Softmax classifier**: model, training objective, gradient update
 - **Classification**: how weights work in this setting
 - **Optimization**: stochastic gradient descent, impact of step size on optimization
 - **Feedforward neural networks**: definition, activation functions
 - **Training neural networks**
 - **Word embeddings**: skip-gram model, definition, properties
 - **Subword tokenization**
 - **Deep averaging networks**: everything from PA1
- **Modern LMs: Background**
 - **Recurrent neural networks**: model, training RNNs,
 - **Sequence to sequence models**: model, training, inference
 - **Sequence to sequence with attention**: applications to machine translation, how attention works, training and inference
- **Modern LMs: Key Ingredients**
 - **Self-attention**: mathematical definition, how it works
 - **Transformers**: architecture details
 - **Transformer language modeling**: everything from PA2, including training, inference, and evaluation (perplexity)
 - **Encoders**: masked language modeling and BERT
 - **Decoders**: causal language modeling, training, inference
 - **Inference in decoders**: nucleus sampling, beam search
 - **Scaling Laws**
 - **Efficient Adaptation**: prompting, ~~parameter-efficient fine-tuning (PEFT)~~

Note Sheet And Electronic Devices

- You are allowed one 8.5" x 11" double-sided note sheet.
- You are **not** allowed calculators or other electronic devices.